Partial fine-tuned neural retriever: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Reduce trainable capacity while keeping much of A0's learned legal representation fixed. The GPU report lists about 28.5 million trainable parameters, so this is a capacity comparison against the full 110.3 million route.

Architecture and mechanism

The same four-window, mean-pool and late-interaction retrieval path is trained with a subset of Phase 1 parameters unfrozen.

Measured result

Scored submission	1039890
Codabench micro-F1@7	0.4028
Archived local metric	0.4036 corrected local-dev micro-F1@7

Corrected local dev micro-F1@7 was 0.4036 while Codabench was 0.4028. The small gap does not identify a clear advantage for either route.

Limitations and next check

The partial tuning boundary is code-defined; do not infer that every layer was equally adapted. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase7_run.py, phase7_export.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.